

Assessment of Village-Level Surface Water Potential in the Ippala Vagu Watershed Region

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Abstract

The management of watersheds requires the planning of surface water resources. For hydrological analysis, runoff estimation is essential. This study aimed to estimate the village level potential surface water storage by combining the water balance components such as the Soil Conservation Service Curve Number (SCS-CN) method for runoff depth assessment, the Rainfall Infiltration Factor (RIF) method for groundwater recharge valuation, the FAO/AGLW method for effective rainfall estimation and rainfall data. Using rainfall data, selected 18 villages within the Ippala Vagu Watershed region performed water balance calculations to evaluate potential surface water storage. In total, 18 villages recorded an annual rainfall volume of 430.50 million cubic meters (MCM), of which 138.24 MCM (32.11%) considered to be effective rainfall. Surface runoff constituted 71.57 MCM (16.62%), while groundwater recharge was estimated at 68.88 MCM (16.00%). The potential surface water storage amounted to 151.82 MCM being 35.27% of the annual precipitation volume. The current storage capacity was merely 73.67 MCM constituting 48.52% of the potential surface water storage. Increasing the number of water harvesting and storage facilities, especially in villages with significant amounts of the potential surface water storage.

1.0 Introduction

The evaluation of surface water availability is a crucial phase for watershed management in rural areas. The measurement of surface runoff is a critical feature in the understanding of water balance components [1]. Watershed size, land use/cover (LULC), surface slope, soil type, and hydro-metrological factors all impact on rainfall-runoff process [2]. Accurate prediction of hydrological events like rainfall-runoff can help rural development and watershed management reduce flood and drought damage to water systems [3]. Watershed modelling requires precise rainfall-runoff simulations for irrigation water management, and water resources planning [4]. Managing and organizing water resources necessitates a comprehensive collection of historical flow data. The calculation of runoff from ungauged or inadequately gauged catchments is a significant difficulty in developing nations like as India, where increased operational and maintenance expenses vary for small and medium rivers. [5]. As a result of the fact that not all watersheds in India are capable of being measured, an indirect method for assessing runoff is required. [6]. Because of its simplicity and effectiveness, the SCS–CN method has gained broad acceptance and has been extensively applied in hydrological studies across India.

The hydrological models are a straightforward presentation of the hydrological system within the watershed. The objective of these models is to elucidate the manner in which a watershed reacts to the hydrological processes that transpire in response to specific inputs [7]. Because of Rainfall-Runoff modelling, the known precipitation data is used to evaluate the unknown runoff data. Several methods are available for modelling rainfall–runoff processes. For the purpose of estimating runoff, the SCS-CN method, which is depended on curve number (CN), is a straightforward and simple approach (USDA 1986). Increasing computer power and wider availability of spatial data allow accurate predictions of runoff. The possibility of fast integration of diverse kind of data in a Geographic Information System (GIS) has led to a considerable expansion in its application in hydrological field [8].

The need for water balance studies under village level within the watershed arises from the growing demand for water and the uneven distribution of rainfall throughout year. The quantity of surface water used by the people is generally very small. It often goes to waste as it eventually drains into the oceans due to the lack of appropriate storage facilities. On one hand, such a huge quantity of good quality water is wasted and on the other, acute water shortage is being felt in several parts of our country. It is therefore, imperative to develop additional storages in our country, to reduce this wastage to as low as possible. In the villages,

public mainly depend on agriculture, which is one of the largest water demand fields. Effective management of water resources is need to be made in the village level to create potential storage, for reducing the existing and prospective water shortage.

2.0 Methodology

The water balance equation contains the relative values of inflow, outflow and change in water storages for selected area.

$$\text{Inflow} - \text{Outflow} = \text{Change in Storage}$$

In study area, the total inflow to the system is precipitation (P) and the total outflow is runoff. The change in total water storage (ΔS) in a selected area may be split up into surface water storage (ΔSS), ground water recharge from rainfall (ΔSG) and moisture storage in soil pores (ΔSM). An analysis of the water balance was conducted to assess the spatial variance of hydrological components in the chosen villages within the Ippala Vagu Watershed region. Potential surface water storage was estimated using the relationship:

$$\text{Potential Surface Water Storage} = \text{Precipitation} - \text{Runoff} - \text{Effective Rainfall} - \text{Groundwater Recharge}$$

The quantity of water available for surface storage was evaluated using a simple water balance method. In this method, total precipitation was main source of water in the system. Effective rainfall denotes the share of precipitation retained within the crop root zone, whereas runoff and groundwater recharge account for water leaving the surface system. Subtracting effective rainfall, runoff and groundwater recharge components from total precipitation reached the remaining portion of the water, and this was considered as the potential surface water storage in the study area. The next sections will provide a description of the estimating methodologies that are utilized for effective rainfall, surface runoff, and groundwater recharge amounts through deep percolation loss.

2.1 Effective Rainfall

The FAO/AGLW method is used to evaluate the effective rainfall, and this method have relation between monthly rainfall and effective rainfall. Effective rainfall provides a practical approximation of the rainfall that is available for crop use after accounting for losses such as runoff, groundwater recharge and potential surface water storage. The empirical equations used in the FAO/AGLW method are given below.

$$P_{eff} = 0.6 \times P - \frac{10}{3} \quad \text{for } P_{month} \leq \frac{70}{3} \text{ mm} \quad \text{eq-1}$$

$$P_{eff} = 0.8 \times P - \frac{24}{3} \quad \text{for } P_{month} > \frac{70}{3} \text{ mm} \quad \text{eq-2}$$

2.2 Surface Runoff

It was estimated using the SCS-CN method. A Digital Elevation Model (DEM) was processed in a ArcGIS software to delineate watershed boundary. LULC and soil maps were integrated to assign Curve Number (CN) values based on land cover values and hydrologic soil groups. These CN values were used in the SCS-CN runoff model to estimate direct surface runoff from daily rainfall, and this model output provided runoff depth.

SCS-CN Method

The SCS-CN method, introduced by the Soil Conservation Service of the United States in 1969, is a widely used conceptual hydrological model for estimating direct runoff generated from daily rainfall depth. This method provides a straightforward, reliable, and consistent approach for runoff assessment. This method is derived from the water balance equation and evaluates rainfall over a specified time interval.

In the SCS-CN method, rainfall (P) is divided into initial abstraction (I_a), actual retention or infiltration (F), and direct runoff (Q).

$$P = I_a + F + Q \quad \text{eq-3}$$

Two other concepts are also used in this method. Equation (4) is based on the initial concept, where S denotes the maximum potential water retention capacity of the soil. This relationship forms the basis for estimating runoff from a given rainfall event.

$$\frac{Q}{(P-I_a)} = \frac{F}{S} \quad \text{eq-4}$$

Equation (5) represents the second concept.

$$I_a = \lambda S \quad \text{eq-5}$$

Combining the above concept,

$$Q = \frac{(P-\lambda S)^2}{P+(1-\lambda)S} \quad \text{for } P > \lambda S \quad \text{eq-6}$$

$$Q = 0 \quad \text{for } P \leq \lambda S \quad \text{eq-7}$$

$$S = \frac{25400}{CN} - 254 \quad \text{eq-8}$$

This CN is determined from LULC, soil type, hydrologic condition, and AMC. Values of ‘ λ ’ varying in the range $0.1 \leq \lambda \leq 0.4$. For use in India conditions $\lambda = 0.1$ to 0.3 . For this study $\lambda = 0.2$ is considered.

Land Use and Land Cover

LULC provide important information on the spatial distribution and characteristics of topography. Accurate LULC classification supports to evaluate the CN values for surface runoff estimation.

Hydrologic Soil Group

Hydrologic Soil Groups (HSGs) classify the soils according to their runoff potential and infiltration capacity. The classification (Table No.1) is based on soil texture, permeability, and drainage characteristics and is commonly divided into four groups (A, B, C, and D). HSG data are widely used in hydrological studies, particularly for runoff estimation.

Table No. 1 Soil texture mapping to hydrologic soil groups [Ref. 13]

Hydrological Soil Group	Soil Textures	Runoff Potential	Infiltration Rate
Group A	Sand, Loamy Sand, Sandy Loam	Low	High
Group B	Loam, Silt Loam, Silt	Moderate Low	Moderate
Group C	Sandy Clay Loam	Moderate High	Low
Group D	Clay Loam, Silty Clay Loam, Sandy Clay, Silty Clay, Clay	High	Very Low

Antecedent Moisture Condition

AMC describes the soil moisture status before a rainfall occurrence and it effects the soil's rate of infiltration and, subsequently, the quantity of surface runoff produced. Dry soils allow greater infiltration and produce less runoff, whereas wet or saturated soils reduce infiltration and increase runoff. The SCS-CN approach categorizes AMC into three types, namely AMC-I (dry), AMC-II (average), and AMC-III (wet). These conditions are determined from the total rainfall received during the five days preceding a storm, with threshold values varying between the growing and dormant seasons (Table No.2). Considering AMC improves the accuracy of runoff estimation by accounting for variations in antecedent soil moisture.

Table No.2 AMC for Determining the value of CN [Ref.26]

AMC Type	Total Rain in previous 5 days	
	Dormant Season	Growing Season
I	Less than 13 mm	Less than 36 mm
II	13 to 28 mm	36 to 53 mm
III	More than 28 mm	More than 53 mm

[Source: K.subramanya]

Runoff Curve Number (CN)

In the SCS–CN method, the CN is the primary parameter used to evaluate surface runoff from rainfall. It reflects the runoff potential of a watershed based on LULC and HSG. Lower CN values represent a stronger infiltration and water storage capacity and hence less runoff, whereas higher CN values present a lower infiltration and higher runoff potential. Therefore, selecting suitable CN values is critical for accurate runoff estimation.

Table No. 3 Runoff Curve Numbers [CN_{II}] for Hydrologic Soil cover Complexes [under AMC-II Conditions]

Sl. No	Land use	Runoff curve numbers for hydrologic soil group			
		A	B	C	D
1.	Water	98	98	98	98
2.	Trees	41	55	69	73
3.	Flooded vegetation	95	95	95	95
4.	Crops	59	69	76	79
5.	Built Area	77	86	91	93
6.	Bare ground	71	80	85	88
7.	Rangeland	68	79	86	89

[Source: K.subramanya]

In order to determine the Curve Number (CN), the study area was first divided into several LULC cover categories and HSG. The weighted contribution (Area × CN) was calculated by multiplying the area of each land use class within each soil group by its standard CN value as specified by Table No.3. After that, the weighted products for each land use class were added together and divided by the watershed's total area. This area-weighted averaging approach provided a representative CN-II for the study area. CN-I and CN-III are calculated using Eq. 9 and 10, respectively.

$$CN_{II} = \frac{\sum(N_i \times A_i)}{A} \quad \text{eq-9}$$

The conversion of CN_{II} to the remaining two AMC states can be performed via the following correlation equations.

$$\text{For AMC-I} \quad CN_I = \frac{CN_{II}}{2.281 - 0.01281CN_{II}} \quad \text{eq-10}$$

$$\text{For AMC-III} \quad CN_{III} = \frac{CN_{II}}{0.427 + 0.00573CN_{II}} \quad \text{eq-11}$$

2.3 Ground Water Recharge

In situations where sufficient groundwater level data are unavailable, groundwater recharge can be evaluated using the RIF method. This method is an empirical technique used to estimate groundwater recharge resulting from rainfall. In this approach, groundwater recharge is determined by multiplying the rainfall infiltration factor (F) by the assessment area (A) and the total seasonal rainfall (P). The recharge (R) is expressed as:

$$R = F \times A \times P \quad \text{eq-12}$$

This approach incorporates a monsoon and non-monsoon rainfall infiltration factor. The recharge contribution from the non-monsoon season is nil if it is less than 10% of the total yearly rainfall. According to the recommendations of the Groundwater Resource Estimation Committee (2015), the rainfall infiltration factor for alluvial regions varies based on geographical location. A value of 22% is suggested for the Indo-Gangetic plains and inland alluvial areas, 16% for the East Coast alluvial regions, and 10% for the West Coast alluvial regions. The study area falls under the East Coast Alluvial Region; therefore, a rainfall infiltration factor of 16% was adopted for this study[14].

2.4 Study Area

The Ippala Vagu watershed encompasses a total of 18 villages distributed across four administrative mandals. Of these, eight villages fall under Vissannapet Mandal (16°56'23.28" N, 80°46'58.08" E), while seven villages are located in A. Konduru Mandal (17°02'52.08" N, 80°05'53.52" E). One village belongs to Reddigudem Mandal. These three mandals are administratively part of the NTR District. The remaining two villages are situated in Chatrai Mandal, which lies within Eluru District. Both NTR and Eluru districts are located in the state of Andhra Pradesh, India.

2.5 Data

Rainfall data used in this study were obtained from the Andhra Pradesh State Development Planning Society (APSDPS), Planning Department, Vijayawada, Andhra Pradesh. The CartoSAT DEM used in this study was acquired from the Bhuvan Geoportal. This high-resolution elevation dataset was derived from stereo imagery captured by the Cartosat-1 satellite mission developed by the Indian Space Research Organization (ISRO). The LULC data used in this study were obtained from the ESRI Global Land Cover dataset, developed by Impact Observatory in collaboration with ESRI. The dataset is derived from 10 m resolution ESA Sentinel-2 Level-2A imagery and provides annual global LULC maps. Soil data for this study were obtained from the Digital Soil Map of the World (DSMW) available through the FAO Soils Portal. The DSMW is a digital version of the FAO–UNESCO Soil Map of the World, providing standardized global soil information in a GIS-compatible format. The dataset was integrated into the GIS environment to support soil characterization and hydrological analysis. The existing storage capacity data were obtained from the Andhra Pradesh Water Resources Information and Management System (APWRIMS) website (<https://apwrims.ap.gov.in>).

3.0 Results and Discussion

The rainfall–runoff model estimates runoff using known precipitation data. Among the commonly used methods, the SCS–CN method is widely used for runoff estimation. It describes the watershed hydrological response considering certain essential elements such as HSG, LULC and AMC.

Figure 1 illustrates the Flow Accumulation Model of the Ippala Vagu Watershed. The model helps identify stream networks, drainage density, and watershed delineation. These identifications useful to take watershed development activities.

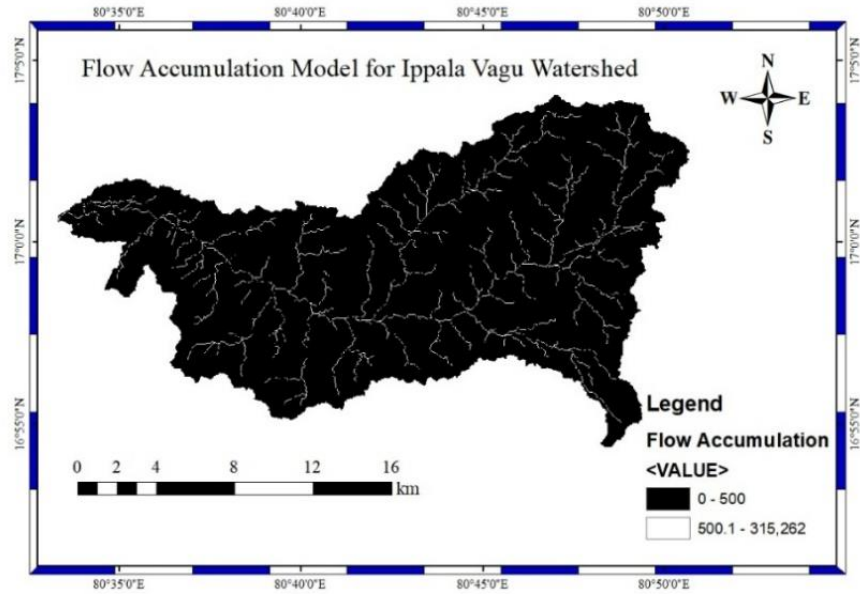


Fig.1 shows flow accumulation model for Ippala Vagu Watershed

Land Use and Land Cover

Figure 2 shows the LULC distribution of the study area. The study region covers 286,498,900 m² in total. Agriculture dominates the region with 227,632,800 m² of Cropland as the major land use class. The built-up area of 18,972,100 m² includes communities and infrastructural development. Tree cover and range land make up natural vegetation, occupying 17,455,200 and 15,877,900 m², respectively. Water bodies occupy 6,493,500 m², but flooded vegetation covers just 64,400 m². The smallest land use category is bare ground, covering only 3,000 m². Cropland is the most common land use, followed by built-up areas, tree cover, rangeland, and water bodies. A tiny portion of the land is occupied by flooded vegetation and bare ground. It is clear that agricultural land is the major land use in the selected study region. The LULC classes provide the basis for assessing CN values to estimate the runoff.

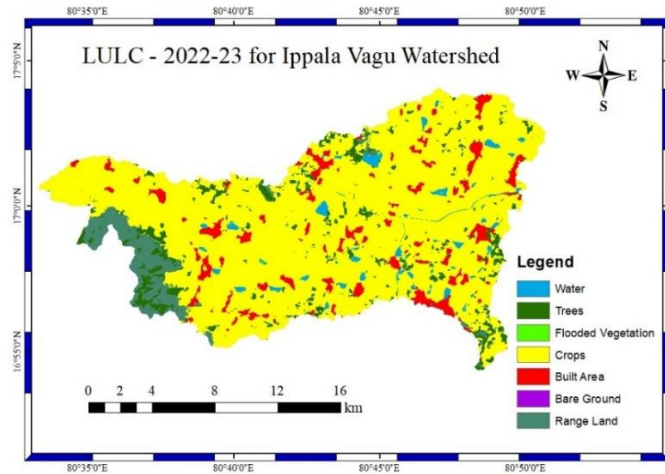


Fig. 2 Land Use Land Cover Map for Ippala Vagu Watershed (2022-23)

Hydrologic Soil Group

Figure 3 displays the soil texture distribution of the Ippala Vagu Watershed. Sandy clay loam (HSG C) dominates the study area, accounting for 80.10% of its total area, while loam (HSG B) inhabits 19.90%. Group C soils exhibit high runoff potential because of their low infiltration capacity, whereas Group B soils allow moderate infiltration and generate relatively less runoff.

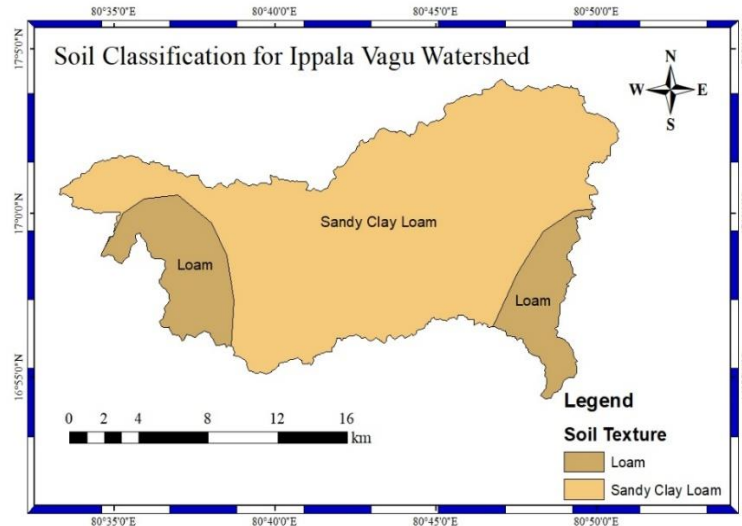


Fig.3 Soil Classification for Ippala Vagu Watershed

The CN was analyzed considering the LULC and soil type which are the elements controlling the infiltration and runoff behavior of the watershed.

CN-I (64.22) demonstrates high infiltration and low runoff potential, CN-II (80.37) represents intermediate runoff conditions and CN-III (90.55) signifies low infiltration and high

runoff potential. The CN values increase indicating a shift from permeable conditions to places producing greater surface runoff.

Fig. 4 compares the monthly rainfall and runoff of the Ippala Vagu watershed region from June to May (2022-23). The results indicate that the rainfall is highly seasonal. The rainfall during the study period was 60 mm in June and reached the peak in July with about 345 mm. Rainfall is still high in August (around 300 mm) and September (around 205 mm) and then gradually decreases to 160 mm in October. Rainfall in November and December fell sharply to less than 30 mm and almost no rainfall is recorded in January and February. Slight increase was seen in March, April and May.

The greatest runoff occurs in July (about 100 mm) and August (around 90 mm) with the months of heaviest rainfall. Runoff drops sharply down in September and October, and is small in the dry months of November to February. Again, some runoff is generated in March and May. Rainfall increases somewhat. The graph shows a solid positive association between the rainfall and runoff in the watershed. Months of high rainfall cause greater runoff whereas months of low rainfall cause little or no runoff.

Marykutty Abraham, M.A et al concluded the annual precipitation for the watershed ranged from 575.7 mm (2002–03) to 3608.0 mm (2005–06). The runoff, as calculated by the NRCS model, ranged from 94.95 mm to 2324.34 mm, with the associated percentages varying from 16.49% to 64.42% for Tank Watershed in Tamil Nadu. In the present study, 17.24% of the total rainfall was converted into runoff. The results of the present study were compared with those reported by Marykutty Abraham, M.A. et al. The SCS Curve Number (CN) method is based on Curve Number values, which directly influence runoff estimation. The Curve Number values adopted for the Ippala Vagu Watershed are lower than those of the tank watershed in Tamil Nadu, resulting in comparatively lower runoff estimates in the present study.

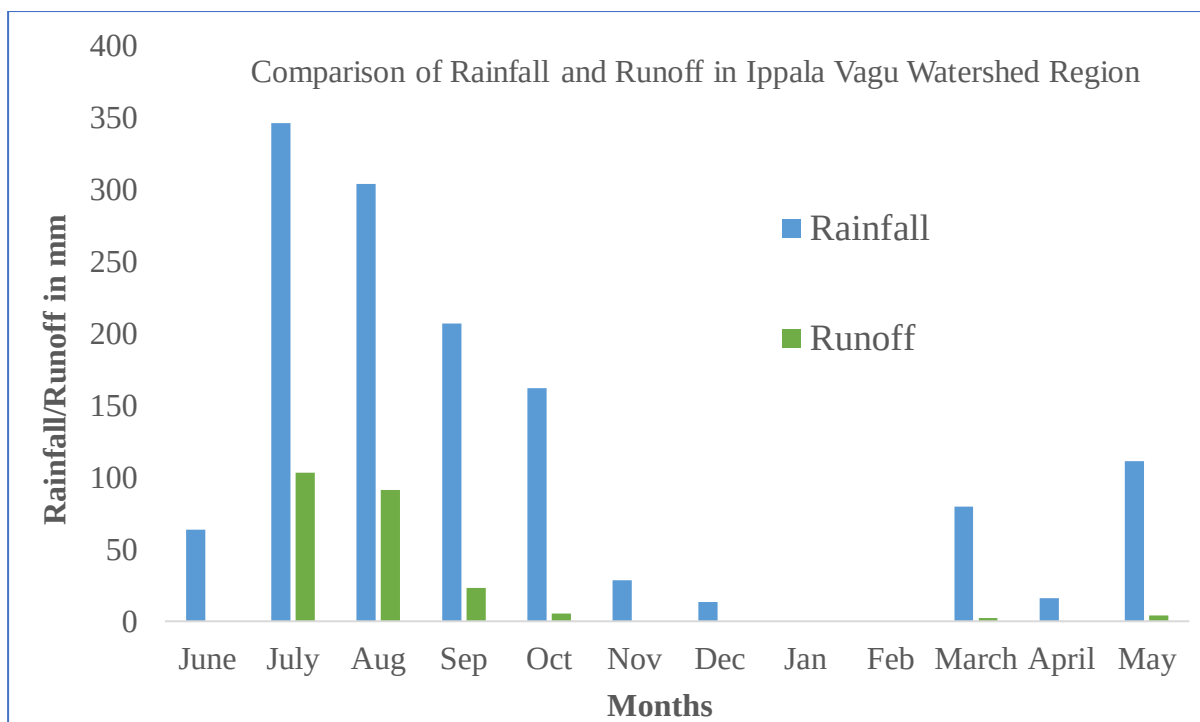


Fig. 4 Comparison of Rainfall and runoff in Ippala Vagu Watershed Region

Figure 5 presents the village-wise water balance components of Ippala Vagu Watershed for the year 2022-23, comprising the volume of rainfall, effective rainfall, runoff, groundwater recharge and potential surface water storage. The volume of each water balance component for each village was estimated by multiplying the village area by the corresponding average water depth. Total rainfall received in the 18 villages of Ippala Vagu watershed was 430.50 million cubic meters (MCM) and effective rainfall was 138.24 MCM. The estimated runoff and groundwater recharge were 71.57 MCM and 68.88 MCM respectively. Total potential surface water storage was 151.82 MCM.

Significant spatial variation is observed among the villages. Putrela recorded the highest rainfall volume (53.61 MCM), followed by Vissannapet (38.13 million m³), Koduru (38.39 MCM), Chanubanda (37.21 MCM), and Chandrupatla (33.70 MCM). These villages also generated relatively high runoff and groundwater recharge due to greater rainfall inputs. In terms of potential surface water storage, Koduru had the highest value (18.86 MCM), closely followed by Chandrupatla (18.03 MCM) Vissannapet (15.45 MCM), Putrela (14.75 MCM) and Chanubanda (11.78 MCM) These locations have greater potential for developing or expanding surface water storage structures.

In contrast, Kummarakuntla recorded the lowest rainfall volume (4.06 MCM) and the least available surface water (0.80 MCM), indicating limited water resource availability. Similarly, Krishnarao Palem, Repudi, Gollamandala, and Kambampadu showed comparatively

lower potential surface water storage, suggesting the need for improved water conservation and storage measures.

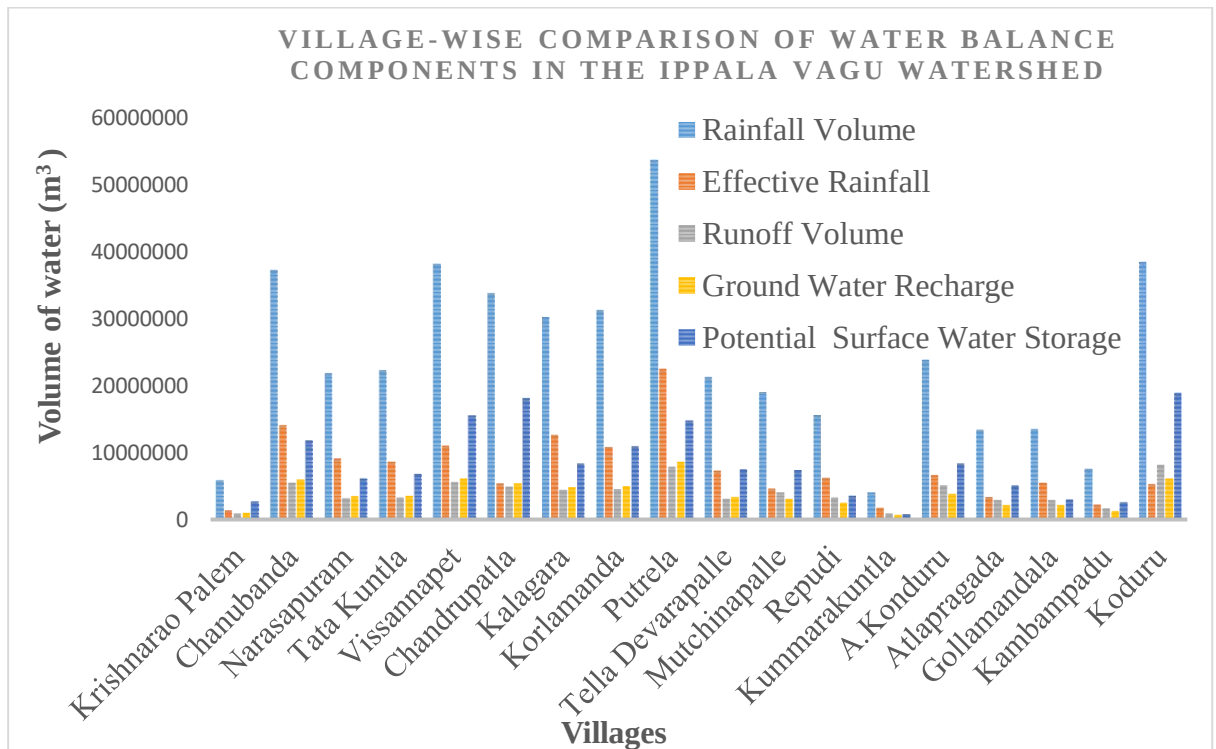


Fig. 5 Village-wise Comparison of water Balance components in the Ippala Vagu Watershed

There is a significant difference between potential surface water storage and storage infrastructure, according to an examination of potential surface water storage and current storage capacity throughout the 18 villages. The current storage capacity is only 73.67 MCM, or roughly 48.5% of the 151.82 MCM of surface water that are available based on estimation. This suggests that a significant volume of surface runoff remains unutilized due to insufficient storage facilities. While Koduru, Chandrupatla, Vissannapet, and Atlapragada have great surface water potential but limited storage, villages like Krishnarao Palem and Kummarakuntla have no storage capacity although having surface water available.

Conversely, villages such as Narasapuram, Tata Kuntla, Kalagara, and Mutchinapalle exhibit storage capacities that are equivalent to or surpass their potential surface water storage, showing good implementation of water conservation plannings. The analysis shows that. the necessity to enhance water storage infrastructure, especially in villages with abundant surface water and insufficient storage capacity, to mitigate runoff losses, expand groundwater recharge, and improve irrigation potential.

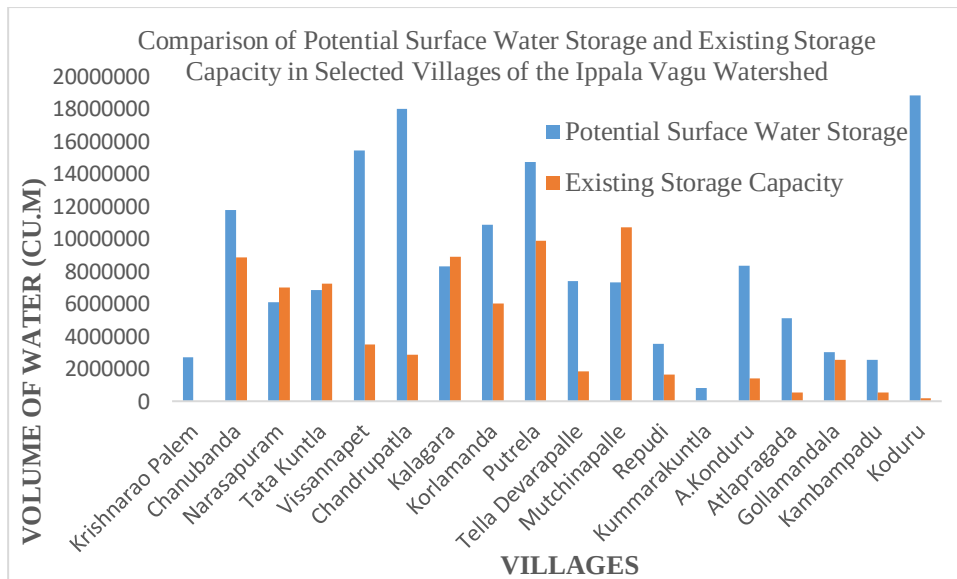


Fig. 6 Comparison of potential surface water storage and existing storage Capacity in selected Village of the Ippala Vagu Watershed

4.0 Conclusion

The study was conducted in the 18 villages of Ippala Vagu watershed. The individual village wise analysis was made for rainfall, effective rainfall, runoff, ground water recharge, potential surface water storage and storage capacity. Individual village results were aggregated later to analyze overall hydrological parameters and water resource potential of the entire watershed basin. The Ippala Vagu Watershed region experienced an annual rainfall volume of 430.50 million m³, with 138.24 million m³ (32.11%) turned into effective rainfall. The projected surface runoff was 71.57 million m³ (16.62%), whereas 68.88 million m³ (16.00%) facilitated groundwater recharge. 151.82 million m³ (35.27% of the total rainfall volume) of potential surface water storage, indicating significant potential for the development of water resources. Thus, increasing water harvesting and storage structures, especially in areas with high runoff and limited storage capacity, will improve water conservation, groundwater recharge, irrigation potential, and sustainable watershed management.

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